Project Title

Eye Disease Detection

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F22BDOCS1M01057

Batch 2022-2026

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A Project presented to

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[Project Documentation]

Version: (1.0) **Date: (05/06/2026)**

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Chapter 1
Introduction

The Eye Disease Detection with Chatbot System is a web-based application developed to help in the early detection and awareness of retinal diseases. The system uses machine learning and image processing techniques to analyze retinal OCT (Optical Coherence Tomography) images and identify common conditions such as CNV, DME, Drusen, as well as normal retinal images. It provides quick and reliable results to help users understand their eye condition.

In addition to disease detection, the system includes an AI-based chatbot that allows users to ask questions, understand their results, and receive intelligent responses and basic information about eye diseases. This makes the application interactive, informative, and easy to use for all types of users.

The system also includes an administrative panel that allows authorized users to monitor system activity, manage user data, and maintain system performance.

This project focuses on creating a simple, user-friendly platform that combines disease detection and guidance in one system, helping to improve awareness and support early diagnosis of retinal conditions.

1.1 Purpose

The purpose of the Eye Disease Detection web application is to provide early detection of retinal diseases such as CNV, DME, and Drusen using machine learning techniques. The system helps users monitor their eye health by analyzing OCT images and identifying possible problems at an early stage. It provides clear results that can assist users in understanding their condition and taking timely action.

The application also supports users by offering basic guidance through an AI-based chatbot that provides intelligent and context-aware responses, helping them learn about eye diseases and make informed decisions about their care. Overall, the system aims to improve awareness, support early diagnosis, and promote better eye health management.

In addition, this Software Requirements Specification (SRS) document describes the overall functionality and requirements of the system. It is intended for developers, designers, project supervisors, and evaluators who will use it to understand, develop, and assess the system effectively.

1.2 Scope

The software product, Eye Disease Detection with Chatbot System, is a web-based application developed to support early awareness of retinal diseases. The system allows users to upload retinal OCT (Optical Coherence Tomography) images and receive automated predictions using a trained machine learning model. It can identify conditions such as CNV, DME, Drusen, as well as normal retinal images. In addition, the system provides guidance through an AI-based chatbot to help users understand their results and receive intelligent, context-aware information about eye diseases.

The application is limited to image-based disease detection and AI-based text guidance through an intelligent chatbot. It does not replace professional medical diagnosis, does not store complete patient records, and does not provide detailed treatment or clinical advice.

The main objective of the system is to promote early detection and improve awareness of retinal diseases by providing a simple and easy-to-use platform for analyzing retinal images.

The system also includes an administrative module that allows authorized users to monitor system activities and manage user-related data.

1.3 Definitions, Acronyms, and Abbreviations.

- **OCT (Optical Coherence Tomography):** A non-invasive imaging technique used to capture high-resolution cross-sectional images of the retina.
- **CNV (Choroidal Neovascularization):** A retinal condition involving abnormal blood vessel growth beneath the retina.
- **DME (Diabetic Macular Edema):** A retinal disease caused by fluid accumulation in the macula due to diabetes.
- **Drusen:** Yellow deposits under the retina, often associated with age-related macular degeneration.
- **AMD (Age-Related Macular Degeneration):** A disease that affects central vision, commonly linked with aging.
- **ML (Machine Learning):** A field of artificial intelligence that enables systems to learn from data and make predictions.
- **CNN (Convolutional Neural Network):** A deep learning model commonly used for image classification tasks.
- **MobileNetV3:** A lightweight deep learning model optimized for efficient image classification, used in this system for retinal image analysis.
- **TensorFlow:** An open-source machine learning framework used for building and training neural network models.
- **Keras:** A high-level API built on TensorFlow, used for designing and training deep learning models.
- **Streamlit:** An open-source Python framework used to develop interactive web applications.
- **Web Application (Web App):** A software application that runs on a web browser.

- **Chatbot:** An AI-based system that interacts with users, processes natural language queries, and provides intelligent, context-aware responses and guidance.
- **Dataset:** A collection of OCT images used to train, validate, and test the machine learning model.
- **Prediction:** The output generated by the system that classifies an OCT image as Normal, CNV, DME, or Drusen.

1.4 References

1. Daniel S. Kermany, et al., (2018), Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning, Cell Journal. (Research paper on retinal disease detection)
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8. IEEE, (2009), *IEEE Standard for Software Design Description (IEEE 1016-2009)* (Standard reference for system design documentation)

1.5 Overview

This Software Requirements Specification (SRS) provides a detailed description of the Eye Disease Detection with Chatbot System, including its functionalities, system features, and requirements. It defines how the system operates and the constraints under which it is developed.

The document is organized to help different users focus on relevant sections. Chapter 2 presents the Software Requirements Specification (SRS), including functional and non-functional requirements of the system. Chapter 3 describes the system design and architecture (SDD), explaining how the system is structured and how its components interact. Chapter 4 covers implementation and testing, including code modules, user interface, and test cases.

1.6 Product Perspective

The Eye Disease Detection with Chatbot System is a web-based application designed for early detection of retinal diseases using OCT images. The system is independent and self-contained, meaning it does not rely on any larger system and can operate on its own through a web browser.

Compared to traditional retinal diagnosis methods, which require manual analysis by medical experts, this system provides automated detection using machine learning techniques. Many existing AI-based solutions require complex installation or specialized software, whereas this system is lightweight, easy to use, and accessible through a browser. It also includes an AI-based chatbot that provides intelligent and context-aware guidance to improve user interaction and understanding of results.

The system interacts with users by allowing them to upload OCT images, process them using a trained model, and display prediction results, while recommendations and additional guidance are provided through the AI-based chatbot.

The system also includes an admin panel for system monitoring, user management, and activity tracking.

Block diagram:

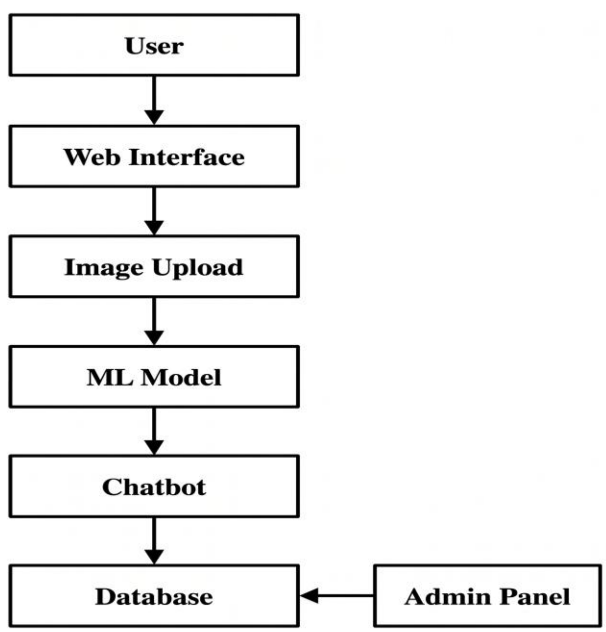

Figure 1.1 : block diagram

1.6.1 System Interfaces

The Eye Disease Detection with Chatbot System operates primarily as an independent web-based application. The system allows users to upload retinal OCT images, which are processed internally by the machine learning model to generate predictions.

The system interacts with an external AI service (Gemini API) to provide chatbot functionality. This interaction enables the system to generate intelligent and context-aware responses to user queries.

All other processing, including image analysis and result generation, is handled within the application itself.

1.6.2 Interfaces

The Eye Disease Detection with Chatbot System provides a graphical user interface (GUI) through a web-based application. Users interact with the system using a web browser, where they can upload retinal OCT images, view prediction results, and communicate with the AI-based chatbot. The interface is designed to be simple, clear, and easy to navigate, allowing users with basic computer knowledge to operate the system without difficulty.

The system includes features such as image upload buttons, result display sections, and chatbot interaction panels. The AI-based chatbot enables users to interact using natural language and receive intelligent, context-aware responses related to retinal diseases and prediction results. The interface is optimized for usability by using clear labels, minimal steps, and responsive design to support different screen sizes such as desktops and mobile devices.

To improve accessibility, the system follows general usability principles such as readable text, simple layout, and intuitive navigation. No specialized training is required to use the system.

In addition to the user interface, the system provides a separate admin interface that allows administrative users to manage system operations, monitor activities, and view system data.

1.6.3 Hardware Interfaces

The Eye Disease Detection with Chatbot System has no direct hardware interface requirements. The system does not interact with or control any specific hardware devices, as it operates as a web-based application. It runs on standard computing devices such as desktops, laptops, or mobile devices using a web browser.

1.6.4 Software Interfaces

The Eye Disease Detection with Chatbot System operates as a web-based application.

The system interacts with an internal database to store and manage user information, uploaded image metadata, prediction results, system logs, and chatbot interaction history. Communication with the database is handled through standard database connection mechanisms.

The system also integrates with an external AI service (Gemini API) to generate intelligent chatbot responses. Communication with the API is performed using standard web-based request and response mechanisms.

All other components, including the user interface and machine learning model, communicate internally within the system.

1.6.5 Communications Interfaces

The Eye Disease Detection with Chatbot System communicates over standard web protocols. The system uses HTTP/HTTPS for communication between the user's web browser and the server hosting the application.

Secure communication is ensured through HTTPS to protect user data, including uploaded OCT images and login credentials. No custom communication protocols are used in this system.

1.6.6 Memory Constraints

The Eye Disease Detection with Chatbot System does not have any strict memory constraints. The system is designed to operate on standard computing devices with typical memory configurations.

However, better performance may be achieved on systems with higher memory capacity, especially during image processing and machine learning prediction tasks. No specific limitations on primary or secondary memory are defined for this system.

1.6.7 Operations

The Eye Disease Detection with Chatbot System operates in an interactive mode where users access the application through a web browser. Users can log in, upload OCT images, view prediction results, and optionally interact with the AI-based chatbot.

Normal operations include user authentication, uploading retinal images, processing them using the machine learning model, and displaying prediction results (such as disease classification and confidence score). The system also allows users to view their previous prediction history after logging in. Admin users can access the admin panel to monitor system activities, view user data, and manage logs.

Special operations include optional interaction with the AI-based chatbot, where users can click the chatbot button to ask questions and receive intelligent, context-aware responses and recommendations related to retinal diseases.

Data processing functions involve image preprocessing (resizing and normalization), prediction using the trained machine learning model, and storing results in the system database. User data, including uploaded images, prediction history, and chatbot interaction history, is stored and remains available for future access after login.
Backup and recovery operations include storing data securely in the database, maintaining logs for system activities, and handling errors to ensure reliable system performance.

1.6.8 Site Adaptation Requirements

The Eye Disease Detection with Chatbot System does not require any special site-specific modifications for operation. The system is a web-based application that can be accessed through standard web browsers on devices such as desktops, laptops, or mobile phones.
For deployment, the system may require installation of necessary software dependencies such as Python and required libraries, as well as database setup for storing user data and prediction history. An internet connection is required if the system is hosted online.
No additional hardware, environmental setup, or specialized configuration is required for users to operate the system.

1.7 Product Functions

The Eye Disease Detection with Chatbot System performs several key functions to assist users in detecting and understanding retinal diseases. The major functions of the system are described below:
The system provides a user interface that allows users to register, log in, and interact with the application through a web browser. Users can upload retinal OCT images using a simple and user-friendly interface.
The system performs disease detection by analyzing uploaded OCT images using a machine learning model. It processes the image and classifies it into one of the categories: CNV, DME, Drusen, or Normal.
After processing, the system provides prediction results, displaying the detected disease along with a confidence score. Users can also view their previous prediction history after logging into the system.
The system includes an optional AI-based chatbot assistance feature, where users can interact using natural language to ask questions and receive intelligent, context-aware explanations or recommendations related to retinal diseases and prediction results.
The system also supports data management functions, including storing user information, uploaded images, and prediction history in a database for future access.
In addition, the system provides an administrative function, where authorized admin users can access an admin panel to monitor system activities, view user data, and manage

prediction history and logs. This helps in maintaining system performance and tracking usage.
Overall, the system integrates image analysis, user interaction, data storage, and administrative monitoring to provide a complete platform for early awareness of eye diseases.

1.8 User Characteristics

The Eye Disease Detection with Chatbot System is designed for a diverse group of users, including medical professionals, students, researchers, and general users. These users may have different levels of knowledge and experience with technology and medical concepts.
Medical professionals such as ophthalmologists and clinicians are expected to have a good understanding of retinal diseases and basic experience with digital tools. They may use the system to quickly analyze OCT images and review results.
Students and researchers may use the system for learning and experimentation purposes. They typically have moderate technical knowledge and are familiar with basic computer operations and web applications.
General users or patients may have limited technical and medical knowledge. Therefore, the system is designed with a simple and user-friendly interface that does not require specialized training. Clear outputs and easy navigation help them understand the results.
Overall, users are expected to have basic computer literacy, including the ability to use a web browser, upload files, and interact with simple interfaces. These characteristics influence the design of the system by emphasizing simplicity, accessibility, and ease of use.
Administrative users (Admins) have higher-level access and are responsible for monitoring system activities, managing user data, and reviewing system logs. They are expected to have basic technical knowledge and familiarity with system operations.

1.9 Constraints

The Eye Disease Detection with Chatbot System is subject to several general constraints that may affect its design and implementation.
The system must follow data privacy and security considerations, as it handles sensitive medical images and user information. Secure communication (HTTPS) and user authentication are required to protect data.
The system operates as a web-based application, so it depends on the availability of a stable internet connection when deployed online. Performance may vary depending on the user's device and network conditions.
The system relies on machine learning models for disease prediction, and the accuracy of results depends on the quality of input OCT images and the training dataset. Therefore,

predictions may not always be fully accurate and should not be considered a substitute for professional medical diagnosis.

There are technical constraints related to the use of specific technologies such as Python-based frameworks and browser compatibility, which may limit support for older systems or unsupported browsers.

The system includes basic audit and control functions, such as user authentication, role-based access (user/admin), and activity logging to monitor system usage and ensure accountability.

In terms of reliability, the system is designed to provide consistent performance; however, it may be affected by server availability, system errors, or resource limitations.

Overall, the system is intended for awareness and educational purposes and is not classified as a critical or safety-dependent medical system.

1.10 Assumptions and Dependencies

The following assumptions and dependencies are considered in the development and operation of the Eye Disease Detection with Chatbot System:

It is assumed that users will provide clear and valid retinal OCT images for accurate disease prediction. Poor-quality or incorrect images may affect the results.

The system assumes that users have access to a stable internet connection when the application is deployed online and are using a modern web browser that supports the system interface.

The system depends on the availability and proper functioning of the machine learning model, which must be correctly trained and loaded to perform predictions.

It is assumed that required software components and dependencies (such as the runtime environment and supporting libraries) are properly installed and configured in the deployment environment.

The system also depends on the availability of a database to store user data, uploaded image metadata, prediction history, and chatbot interaction history.

For chatbot functionality, it is assumed that the AI-based chatbot can process user queries and generate intelligent responses based on its model and available knowledge. However, responses may vary depending on the input and may not always be fully accurate or cover all possible cases.

Any changes in these assumptions, such as lack of internet access, unsupported browser, missing dependencies, or incorrect model configuration, may affect the system’s functionality and require updates to the system requirements.

1.11 Apportioning of Requirements

The Eye Disease Detection with Chatbot System includes all core functionalities in the current version, such as user authentication, OCT image upload, disease prediction, result

display, chatbot interaction, and basic admin panel features. These functions are essential for the system to operate effectively and provide early awareness of retinal diseases.
Some additional features are planned for future versions of the system. These include improving the chatbot to provide more advanced and intelligent responses, extending the system to detect additional eye diseases, and enhancing the user interface for better usability and experience. Advanced features such as detailed data analysis, reporting tools, and integration with external medical systems may also be considered in future updates.
This division allows the system to deliver its main functionality in the current version while providing flexibility for further improvements and expansion.

Chapter 2

Software Requirements Specification

2.1 Specific Requirements

This section defines the detailed requirements of the Eye Disease Detection with Chatbot System, including all system inputs, outputs, and functions in a form that supports system design and testing.

The system accepts inputs such as user registration data (name, email, password), login credentials, retinal OCT images, and user queries for the AI-based chatbot. Based on these inputs, the system processes data and generates outputs including disease prediction results (CNV, DME, Drusen, or Normal), confidence scores, chatbot responses through the Gemini API, and stored user history.

All system functions are externally observable. These include user authentication, image upload, disease prediction using a machine learning model, chatbot interaction through an external AI service, data storage, and administrative operations such as monitoring system activity and managing user data.

Each requirement is clearly defined, complete, consistent, and testable. Requirements are uniquely identified to ensure traceability and are organized in a structured manner for clarity and understanding.

2.2 External Interfaces

This section describes all inputs to and outputs from the Eye Disease Detection with Chatbot System, including their purpose, format, source, and relationships.

2.2.1 User Registration Input

Name of item: User Registration Data
Purpose: To create a new user account
Source: User
Data format: Text (name, email, password)
Valid range: Valid email format, password with required length
Output: Account creation confirmation or error message
Screen format: Registration form

2.2.2 User Login Input

Name of item: Login Credentials
Purpose: To authenticate user access
Source: User
Data format: Email and password

Valid range: Must match registered data
Output: Login success or error message
Screen format: Login interface

2.2.3 Image Upload Input

Name of item: Retinal OCT Image
Purpose: To perform disease detection
Source: User
Data format: JPG/PNG image
Valid range: Valid image file only
Output: Image accepted or error message
Screen format: Upload interface

2.2.4 Prediction Output

Name of item: Disease Prediction Result
Purpose: To display detected disease
Destination: User
Data format: Text (CNV, DME, Drusen, Normal)
Units: Not applicable
Timing: Generated after processing image
Output format: Result display section

2.2.5 Chatbot Input

Name of item: User Query
Purpose: To request information or guidance
Source: User
Data format: Text input
Output: AI-generated response
Interface: Chat window

2.2.6 Chatbot Output

Name of item: Chatbot Response
Purpose: To provide information and guidance
Destination: User
Source: Gemini API
Data format: Text response
Timing: Real-time response
Interface: Chat window

2.2.7 Admin Interface

Name of item: Admin Data Access
Purpose: To manage users and system logs
Source: Admin user
Output: User data, logs, system activity
Screen format: Admin dashboard

2.3 Functions

This section defines the functional requirements of the Eye Disease Detection with Chatbot System. These requirements describe the actions performed by the system in processing inputs and generating outputs.

2.3.1 User Authentication

- The system shall allow users to register using name, email, and password.
- The system shall allow users to log in using valid credentials.
- The system shall validate user input during registration and login.
- The system shall display error messages for invalid credentials.

2.3.2 Image Upload and Validation

- The system shall allow users to upload retinal OCT images.
- The system shall validate the uploaded image format (JPG/PNG).
- The system shall reject invalid or corrupted images.

2.3.3 Disease Detection

- The system shall process uploaded images using a machine learning model.
- The system shall classify the image as CNV, DME, Drusen, or Normal.
- The system shall generate a confidence score for the prediction.

2.3.4 Result Generation

- The system shall display prediction results to the user.
- The system shall store prediction results in the database.
- The system shall allow users to view their prediction history.

2.3.5 Chatbot Function

- The system shall allow users to enter queries in the chatbot interface.
- The system shall send user queries to the chatbot service.
- The system shall receive responses from the chatbot and display them to the user.
- The system shall store chatbot interaction history.

2.3.6 Admin Functions

- The system shall allow admin login.
- The system shall allow admin to view user data.
- The system shall allow admin to monitor system activity.
- The system shall allow admin to access system logs.

2.3.7 Error Handling

- The system shall display error messages for invalid inputs.
- The system shall handle system failures without crashing.
- The system shall ensure data is not lost during errors.

2.4 Performance Requirements

This section defines the performance requirements of the Eye Disease Detection with Chatbot System in measurable terms.

2.4.1 Static Requirements

- The system shall support multiple users accessing the system simultaneously.
- The system shall handle storage of user data, uploaded image metadata, prediction results, and chatbot interaction history.
- The system shall support standard web browsers for user access.

2.4.2 Dynamic Requirements

- The system shall process image predictions within 2–5 seconds under normal conditions.
- The system shall generate chatbot responses within 3–6 seconds.
- The system shall handle multiple user requests without significant delay.
- The system shall retrieve stored data (history, logs) within 2 seconds.

2.5 Logical Database Requirements

This section defines the logical requirements for the data stored and managed by the Eye Disease Detection with Chatbot System.

2.5.1 Data Types

The system shall store the following types of information:

- User information (name, email, password)
- Uploaded image metadata (file name, upload time)
- Prediction results (disease type, confidence score)
- Chatbot interaction history (user query, response)
- System logs (user activity, errors)

2.5.2 Data Access

- The system shall allow users to access their prediction history and chatbot history.
- The system shall allow admin users to access all system data and logs.
- The system shall restrict unauthorized access to data.

2.5.3 Data Relationships

- Each user shall be associated with multiple prediction records.
- Each user shall be associated with multiple chatbot interactions.
- Prediction records shall be linked to uploaded images.

2.5.4 Data Integrity

- The system shall ensure that user data is valid and consistent.
- The system shall prevent duplicate user accounts.
- The system shall maintain accurate relationships between data entities.

2.5.5 Data Retention

- The system shall store user data, prediction history, and chatbot interactions for future access.
- The system shall maintain logs for monitoring and analysis.

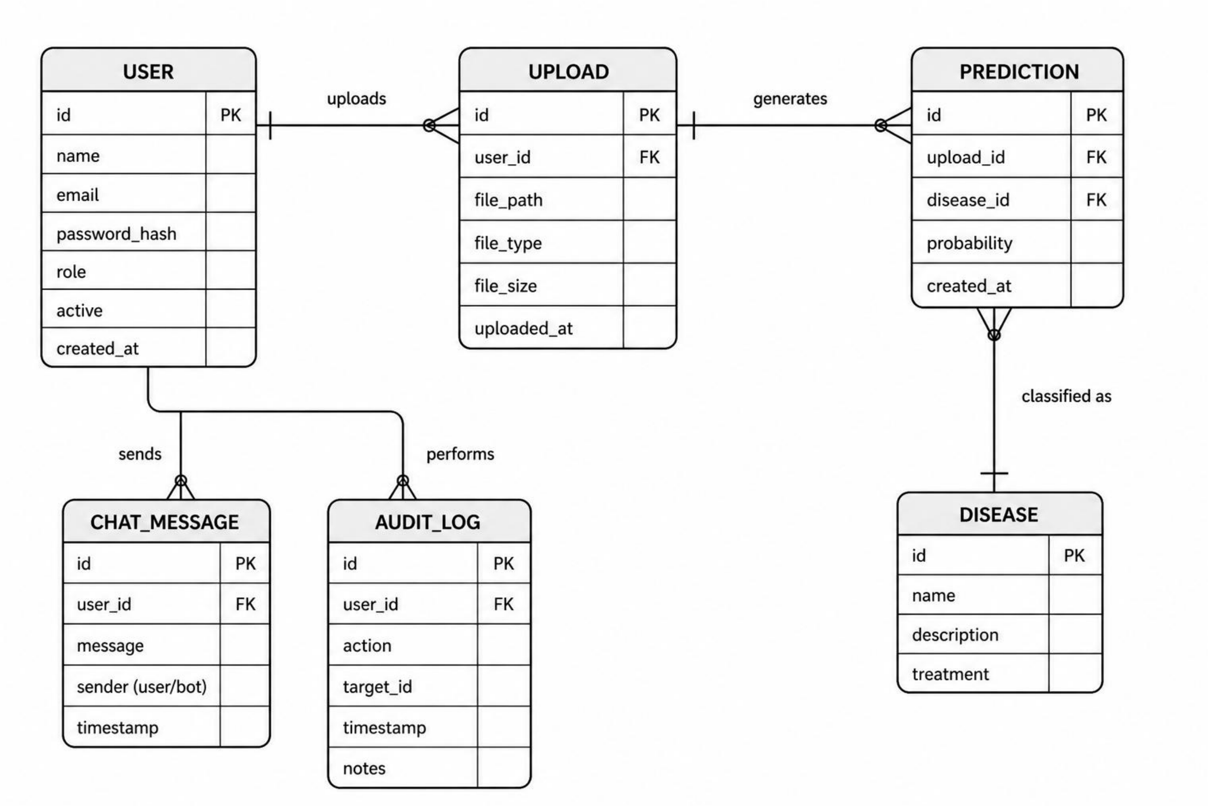

Figure 2.2: Entity Relationship Diagram of Eye Disease Detection System with Chatbot System

2.6 Design Constraints

The following design constraints apply to the Eye Disease Detection with Chatbot System:

- The system shall be implemented as a web-based application accessible through standard web browsers.
- The system shall use a machine learning model for retinal disease detection.
- The system shall integrate with an external chatbot service (Gemini API) for generating responses.
- The system shall store data in SQLite, a lightweight relational database management system, for managing user information, prediction results, chatbot history, and system logs.
- The system shall operate on systems with standard computing resources without requiring specialized hardware.
- The system shall support commonly used image formats such as JPG and PNG.
- The system shall depend on internet connectivity for chatbot functionality.
- The system shall use SQLite as an embedded database, requiring no separate database server installation or configuration.

2.6.1 Standards Compliance

The Eye Disease Detection with Chatbot System shall comply with the following standards and conventions:

- The system shall follow a consistent report format for displaying prediction results and system outputs.
- The system shall use clear and standardized naming conventions for data fields and SQLite database attributes.
- The system shall maintain audit logs to record user and admin activities for tracking and monitoring purposes.
- The system shall ensure data integrity by maintaining accurate and consistent records in the SQLite database.
- The system shall follow basic web application security practices, including user authentication and controlled access.
- The system shall use SQLite's built-in ACID (Atomicity, Consistency, Isolation, Durability) properties to ensure reliable and consistent data transactions.

2.7 Software System Attributes

This section defines the non-functional requirements of the Eye Disease Detection with Chatbot System. These attributes describe the quality characteristics that the system must satisfy.

2.7.1 Reliability

- The system shall provide consistent prediction results for valid OCT images.
- The system shall handle invalid inputs and system errors without failure.
- The system shall ensure that data is not lost during processing or system errors.

2.7.2 Availability

- The system shall be available to users whenever required through a web browser.
- The system shall recover from failures without affecting stored data.
- The system shall allow users to continue operation after system restart.

2.7.3 Security

- The system shall require user authentication for access.
- The system shall restrict admin functionalities to authorized users only.
- The system shall protect user data from unauthorized access.
- The system shall maintain logs of user and admin activities.

2.7.4 Maintainability

- The system shall be structured in a modular way for easy updates and maintenance.
- The system shall allow modifications without affecting other system components.

2.7.5 Portability

- The system shall operate on different operating systems through a web browser.
- The system shall support commonly used web browsers such as Chrome, Firefox, and Edge.
- The system shall be accessible on different devices including desktops, laptops, and mobile devices.
- The system shall not depend on platform-specific hardware or software.

2.8 Organizing the Specific Requirements

The requirements of the Eye Disease Detection with Chatbot System are organized to improve clarity, readability, and implementation.

2.8.1 System Mode

The system operates in an interactive web-based mode where users perform actions such as registration, login, image upload, and chatbot interaction. The system processes these inputs and provides corresponding outputs such as prediction results and chatbot responses in real time.

2.8.2 User Class

The system supports two main user classes:

- **General Users:** Users who can register, log in, upload OCT images, view prediction results, and interact with the chatbot.
- **Admin Users:** Users who can access the admin panel to monitor system activity, manage user data, and view system logs.

2.8.3 Objects

The system is organized around key objects such as user, uploaded image, prediction, chatbot interaction, and system logs.

2.8.4 Feature

A feature is an externally visible service provided by the system. The system is organized based on key features such as user authentication, image upload, disease detection, chatbot interaction, and administrative functions.

2.8.5 Stimulus

Some system functions are organized based on stimuli. In this system, stimuli include user actions such as registration, login, image upload, and chatbot queries, which trigger corresponding system operations.

2.8.6 Response

Some system functions are organized based on responses generated by the system. In this system, responses include prediction results, chatbot replies, confirmation messages, and error messages produced in response to user actions.

2.8.7 Functional Hierarchy

The system functions are organized in a hierarchical structure based on major operations. High-level functions include user management, image processing, disease detection, chatbot interaction, data management, and administrative operations. These functions are further divided into sub-functions based on inputs, outputs, and data handling within the system.

2.9 Additional Comments

The requirements of the Eye Disease Detection with Chatbot System are organized using a combination of feature-based and user-based approaches to improve clarity and understanding.
Standard documentation techniques such as data flow diagrams and entity relationship diagrams are used to represent system processes and data relationships.
All functional requirements are described in clear and structured form to support system design, implementation, and testing.

Chapter 3
System Design and Architecture

3.1 Architectural Design

The Eye Disease Detection with Chatbot System is designed using a modular architecture where the system is divided into high-level subsystems, each responsible for a specific functionality. This modular structure improves maintainability, scalability, and clarity of the system design.

The system is divided into the following major subsystems:

- **User Interface Module**

  Handles user interactions such as registration, login, image upload, result display, and chatbot communication through a web interface.

- **Authentication Module**

  Manages user and admin authentication to ensure secure access to the system.

- **Image Processing & Prediction Module**

  Processes uploaded OCT images, performs preprocessing, and uses a machine learning model to classify diseases (CNV, DME, Drusen, Normal).

- **Database Module**

  Stores and manages user data, uploaded image metadata, prediction results, chatbot interaction history, and system logs.

- **Chatbot Module**

  Handles user queries and communicates with the external AI service (Gemini API) to generate responses.

- **Admin Module**

  Allows admins to monitor system activity, manage users, and view logs.

Subsystem Interaction

These subsystems collaborate to achieve the overall functionality of the system:

- The User Interface Module interacts with users and sends requests to other modules.
- The Authentication Module verifies user credentials before granting access.
- The Image Processing Module receives uploaded images and returns prediction results.

- The Chatbot Module processes user queries and returns responses.
- The Database Module stores and retrieves all system data.
- The Admin Module accesses the database to monitor and manage system operations.

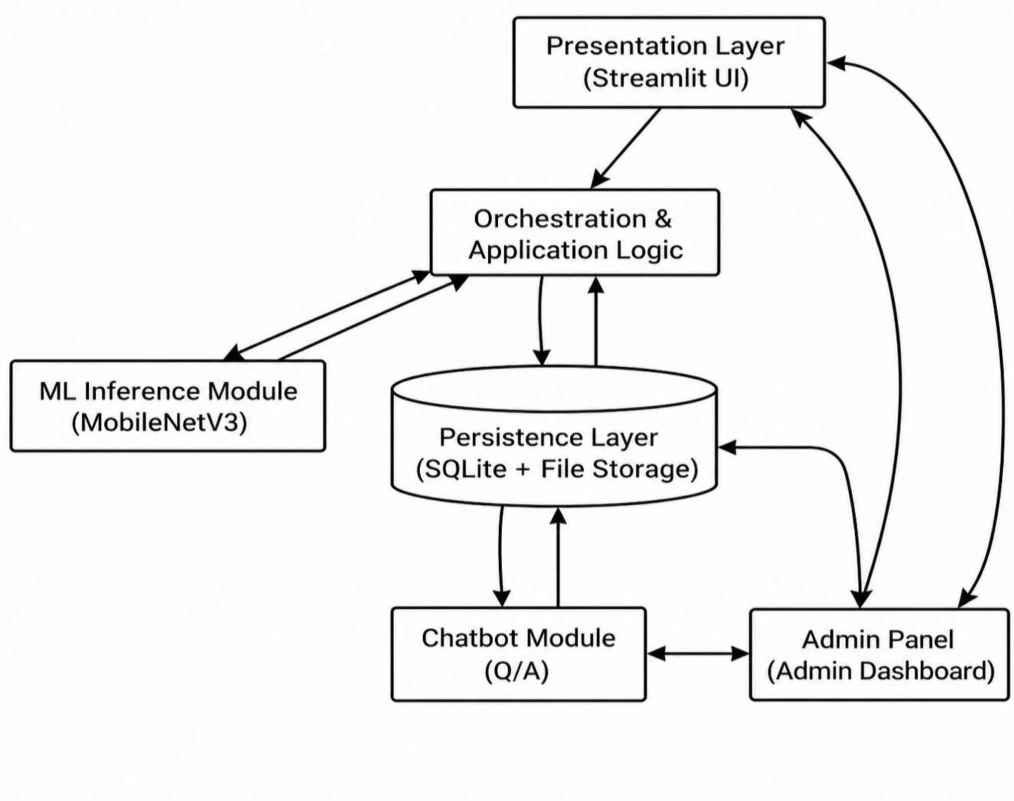

Figure 3.1: System Architecture Diagram of Eye Disease Detection with Chatbot System

3.2 Decomposition Description

The system is decomposed using a functional approach, where each subsystem is responsible for a specific task. This helps in understanding how the overall system is structured and how different components interact to achieve the required functionality.

3.2.1 Top-Level Data Flow Diagram (DFD – Level 0)

The Level-0 Data Flow Diagram represents the Eye Disease Detection with Chatbot System as a single process and shows the interaction between external entities and the system.

- Users provide inputs in the form of OCT images and chatbot queries.
- The system processes images using the machine learning module and generates prediction results.

- The system communicates with the chatbot module to generate responses using the Gemini API.
- All data including prediction results, chatbot interactions, and logs are stored in the database.
- Admin users access the system to monitor activities and retrieve stored data.

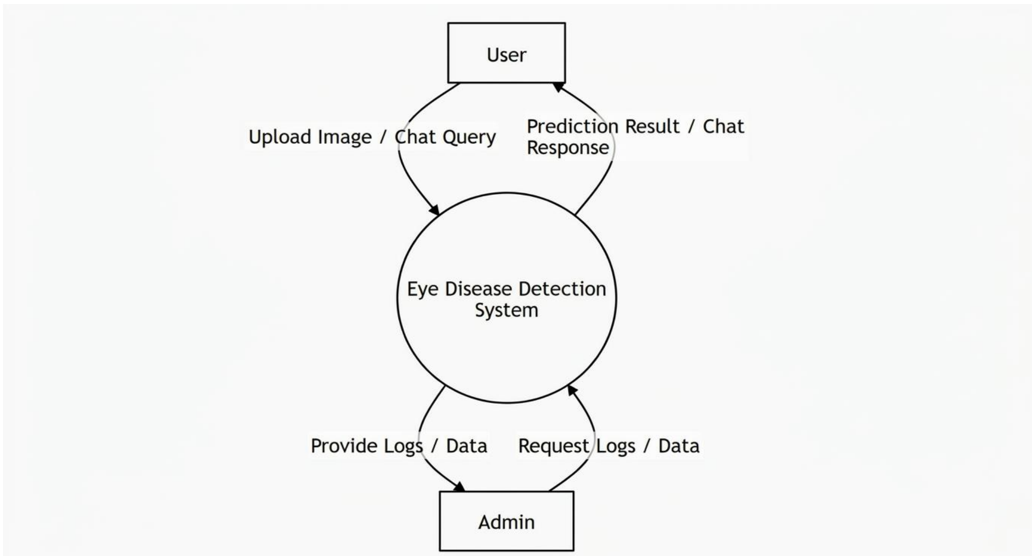

Figure 3.2.1: DFD Level 0 – Eye Disease Detection System

3.2.2 Functional Decomposition of Subsystems

The system is divided into the following functional subsystems:

A. Presentation Layer (UI)

- Handles user and admin interaction with the system.
- Provides interfaces for login, image upload, prediction display, chatbot interaction, and admin dashboard.

B. Application Logic Layer

- Manages system workflow and request processing.
- Handles user requests such as image upload, prediction, chatbot queries, and data retrieval.
- Controls access for users and admin roles.

C. ML Inference Module

- Performs preprocessing of OCT images.
- Uses the trained MobileNetV3 model to classify diseases (CNV, DME, Drusen, Normal).

- Generates prediction results with confidence scores.

D. Chatbot Module

- Handles user queries through the chatbot interface.
- Sends queries to the external AI service (Gemini API).
- Receives and displays responses to users.
- Stores chatbot interaction history in the database.

E. Persistence Layer (Database)

- Stores user data, uploaded image metadata, prediction results, chatbot history, and system logs.
- Provides data access for both users and admin.

F. Admin Module

- Allows admin users to monitor system activity.
- Provides access to user data, prediction history, and system logs.

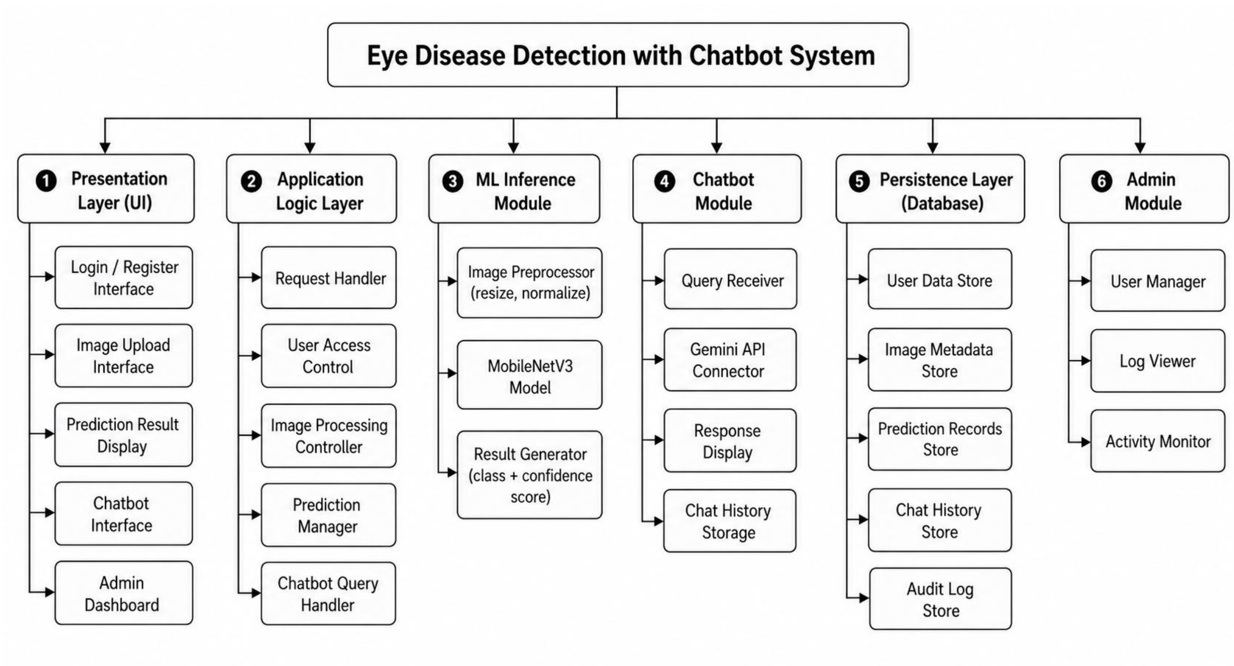

Figure 3.2.2 : System Architecture of Eye Disease Detection with Chatbot Integration

3.2.3 Structured Decomposition Diagram

The structured decomposition diagram shows the internal breakdown of each subsystem into smaller functional components. It represents how the system is organized into layers and modules for better maintainability and scalability.

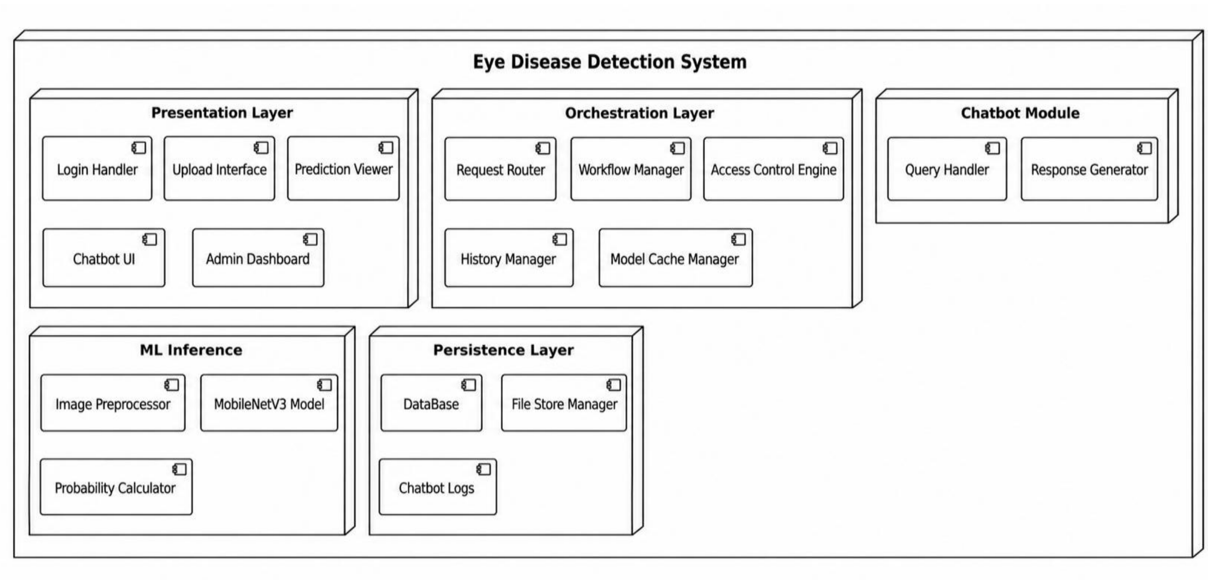

Figure 3.2.3 Structured Decomposition Diagram

3.3 Design Rationale

The Eye Disease Detection with Chatbot System uses a modular layered architecture to organize system components into separate functional units such as user interface, application logic, machine learning module, chatbot module, and database.

This architecture was selected because it improves maintainability and flexibility. Each module performs a specific task, making it easier to update or modify individual components without affecting the entire system. For example, the machine learning model or chatbot functionality can be updated independently.

The system also integrates a chatbot module using the Gemini API, which allows intelligent responses without increasing internal system complexity. The database layer ensures proper storage of user data, prediction results, chatbot history, and system logs. The admin module provides monitoring and control features for system management.

A monolithic architecture was considered but not selected because it makes the system difficult to maintain and scale. A fully distributed architecture was also not used due to its complexity for this project.

One trade-off in this design is the dependency on internet connectivity for chatbot functionality. However, the selected architecture provides a good balance between simplicity, performance, and scalability.

3.4 Data Description

The Eye Disease Detection with Chatbot System transforms input data such as user information, retinal OCT images, and chatbot queries into structured data stored and processed within the system.

The system uses a database to organize and manage all major data entities. These include user data, uploaded image metadata, prediction results, chatbot interaction history, and system logs. Each entity is stored in structured form and linked through relationships such as user ID and upload ID.

User data includes basic details such as name, email, and password. Uploaded images are stored as files, while their metadata (file name, upload time, and user reference) is stored in the database. The machine learning module processes image data and generates prediction results, including disease type and confidence score, which are also stored.

The chatbot module processes user queries and stores both the query and response as part of chatbot interaction history. The admin module accesses stored data to monitor system activity and manage records.

Overall, the data is organized in a structured and relational manner, ensuring efficient storage, retrieval, and processing across different system modules.

3.5 Data Dictionary

The following data elements are used in the Eye Disease Detection with Chatbot System:

- **User (Entity):** Stores user information. Fields: id (int), name (string), email (string), password_hash (string), role (string).
- **Upload (Entity):** Stores uploaded image details. Fields: id (int), user_id (int), filename (string), upload_time (datetime).
- **Prediction (Entity):** Stores disease prediction results. Fields: id (int), upload_id (int), user_id (int), predicted_class (string), probability (float), timestamp (datetime).
- **ChatHistory (Entity):** Stores chatbot interactions. Fields: id (int), user_id (int), query (string), response (string), timestamp (datetime).
- **AuditLog (Entity):** Stores admin activity logs. Fields: id (int), admin_id (int), action (string), timestamp (datetime).

3.6 Component Design

This section describes the internal working of major system components using simplified pseudo code.

1. User Authentication

```
function login(email, password):
    if email exists in database:
        if password matches:
            allow access
        else:
            show error
```

```
        else:
    show error
```

2. Image Upload and Prediction

```
        function upload_image(image):
            validate image format
            preprocess image
            result = model.predict(image)
            store result in database
    return result
```

3. Chatbot Processing

```
        function chatbot_query(query):
            send query to Gemini API
            receive response
            store query and response
    return response
```

4. Data Storage

```
        function save_data(data):
            connect to database
            store data
    confirm save
```

5. Admin Operations

```
        function view_logs():
            retrieve logs from database
            display logs
        function manage_users():
            retrieve user data
    perform update/delete actions
```

Chapter 4
Implementation and Testing

4.1 Code Modules

The Eye Disease Detection with Chatbot System is implemented using modular code structure, where each module performs a specific function.
The main code modules include:

- **User Module:** Handles user registration, login, and session management.
- **Admin Module:** Provides functionalities for managing users, viewing logs, and monitoring system activity.
- **Image Processing Module:** Handles image validation, preprocessing, and preparation for prediction.
- **Prediction Module:** Uses the trained machine learning model to classify OCT images and generate results.
- **Chatbot Module:** Processes user queries and communicates with the Gemini API to generate responses.
- **Database Module:** Manages storage and retrieval of user data, prediction results, chatbot history, and logs.

4.2 Database Connectivity

The system connects to a database to store and manage all required data. The database is used to store user information, uploaded image metadata, prediction results, chatbot interaction history, and system logs.
Database connectivity is established using standard database connection methods. The application performs operations such as inserting new records, retrieving stored data, and updating user or system information. All modules interact with the database through defined functions to ensure data consistency and integrity.

4.3 Overview of User Interface

The system provides a simple and user-friendly web-based interface that allows users to interact with all features easily.
Users can register and log in to the system. After login, users can upload retinal OCT images and receive prediction results showing the detected disease and confidence score. Users can also interact with the AI-based chatbot to ask questions and receive guidance related to eye diseases.
The system provides clear feedback such as success messages, error messages, prediction results, and chatbot responses. Admin users can access a separate dashboard to monitor system activity, view user data, and manage logs.

4.4 Screen Images

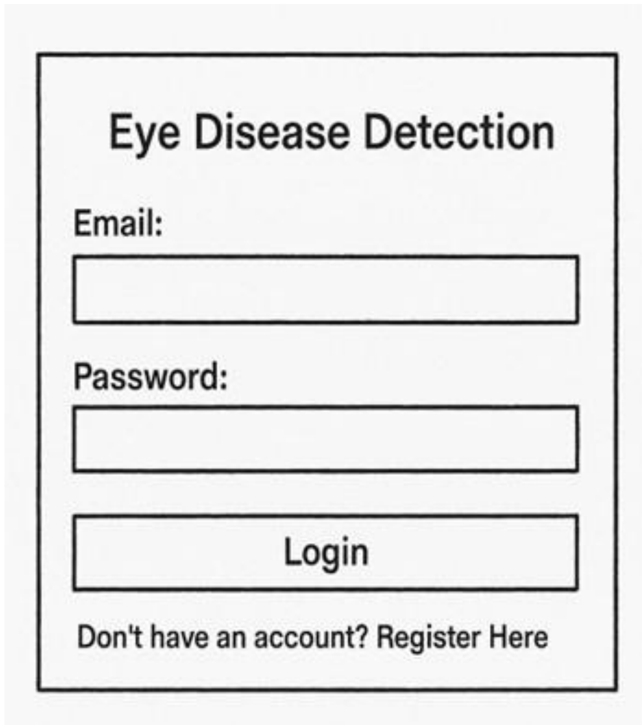

Figure 4.1: Login/registration Interface of the System

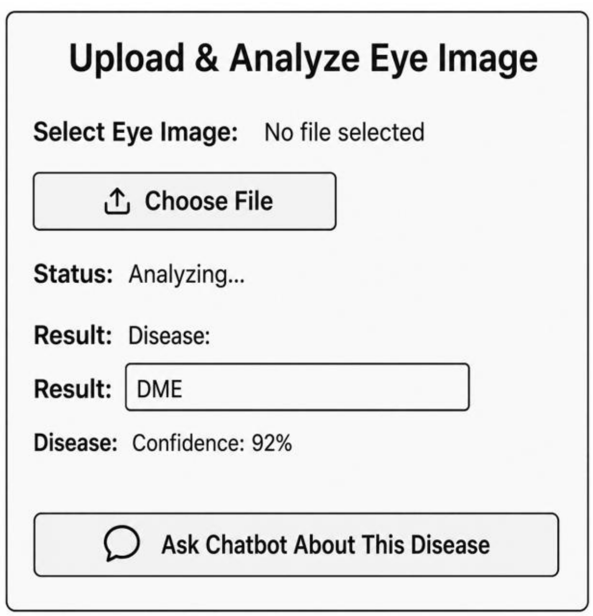

Figure 4.2: Image Upload and Disease Detection Interface

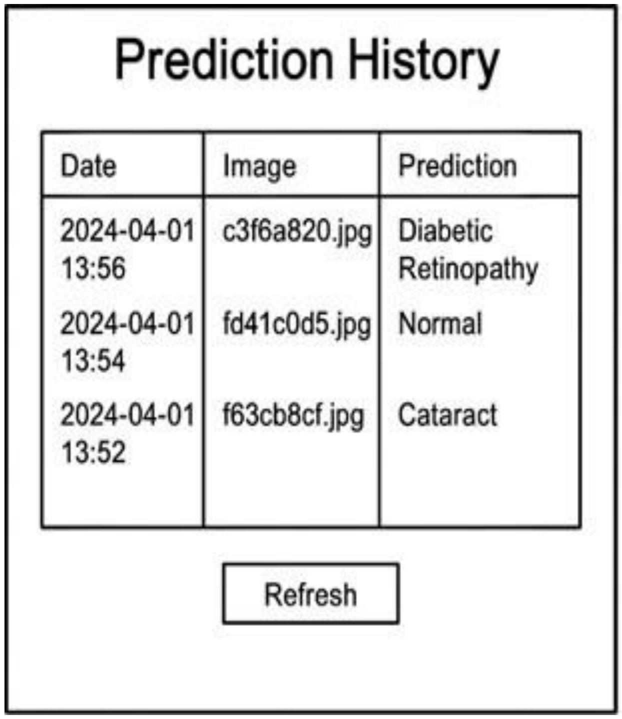

Figure 4.3: Prediction History Interface

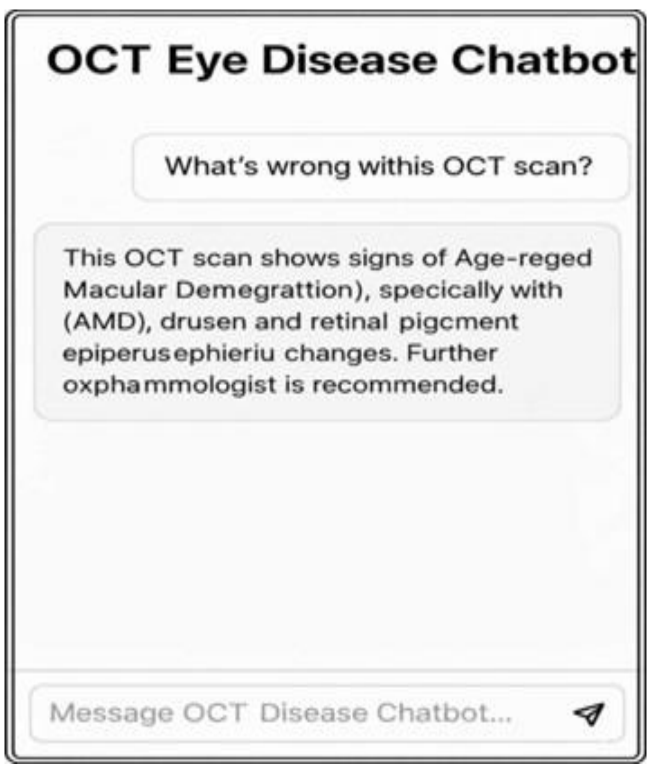

Figure 4.4: Chatbot Interaction Interface

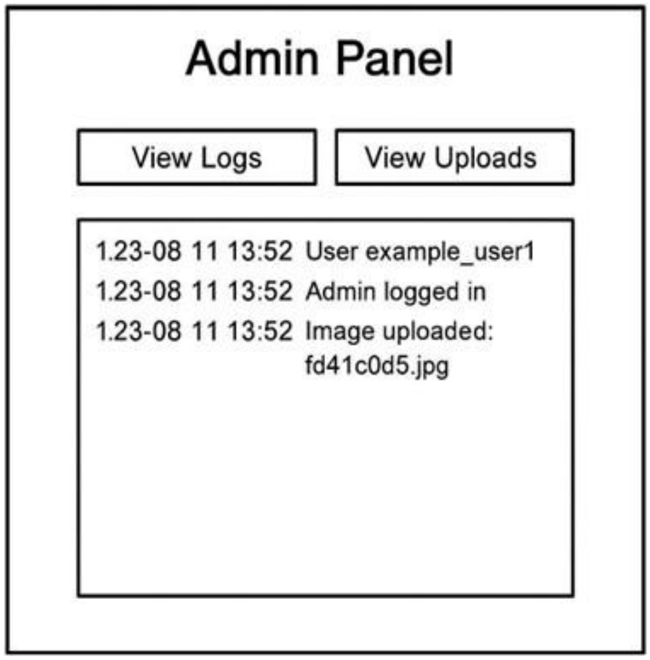

Figure 4.5: Admin Panel Interface

4.5 Screen Objects and Actions

The system interface consists of various screen objects that allow users and admin to interact with the application. Each object performs a specific action when used.

• **Text Fields:**
Used to enter user information such as name, email, password, and chatbot queries. These fields accept user input and send data to the system for processing.

• **Buttons:**
Used to perform actions such as Register, Login, Upload Image, Submit Query, and Logout.

• **File Upload Control:**
Allows users to select and upload retinal OCT images. The system validates the file format before processing.

• **Display Area:**
Displays outputs such as prediction results (disease type and confidence score), chatbot responses, and system messages.

• **Chat Window:**
Provides an interface for users to communicate with the AI-based chatbot. It displays user queries and system responses.

• **Navigation Menu:**
Allows users to move between different pages such as login, upload, results, chatbot, and admin dashboard.

• **Admin Dashboard Controls:**
Includes options for viewing user data, prediction history, and system logs. Admin can perform actions such as monitoring and managing system data.

Each screen object is designed to provide clear interaction and immediate feedback to the user, ensuring ease of use and efficient system operation.

4.6 Test Cases

The system is tested using different test cases to ensure that all functionalities work correctly under normal and abnormal conditions.

Table: Test Cases

Test Case ID	Function	Input	Expected Output
TC-01	User Registration	Valid name, email, password	Account created successfully
TC-02	User Login	Correct email and password	Login successful
TC-03	User Login	Incorrect password	Error message displayed
TC-04	Image Upload	Valid OCT image (JPG/PNG)	Image accepted successfully
TC-05	Image Upload	Invalid file format	Error message displayed
TC-06	Disease Prediction	Uploaded image	Disease type with confidence score
TC-07	Chatbot Query	User question	AI-generated response displayed
TC-08	Chatbot Query	Empty input	Error or no response message
TC-09	View History	Logged-in user	Display of previous predictions
TC-10	Admin Login	Valid admin credentials	Admin dashboard displayed
TC-11	Admin Action	View logs/users	Data displayed correctly
TC-12	System Error Handling	System failure or invalid input	Appropriate error message